



Zhen Lu

PH.D. OF EPIDEMIOLOGY AND BIostatISTICS · SUN YAT-SEN UNIVERSITY

Shanghai, China

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Education

Sun Yat-sen University

PH.D. OF EPIDEMIOLOGY AND BIostatISTICS

Shenzhen, China

Sep. 2022 — Jun. 2026

- Advisor: [Siyang Liu](#), [Huachun Zou](#)
- Research Fields: Clinical Epidemiology, Machine Learning, Comparative Effectiveness, Statistical Genetics

Harbin Medical University

M.Sc. OF EPIDEMIOLOGY AND BIostatISTICS

Harbin, China

Sep. 2018 — Jun. 2021

- Advisor: Yan Liu
- Research Fields: Biostatistics, Data Science

Harbin Medical University

B.Sc. OF PREVENTIVE MEDICINE

Harbin, China

Sep. 2013 — Jun. 2018

- Courses: Epidemiology, Health Statistics

Sun Yat-sen University

INTERNATIONAL SUMMER SCHOOL

Shenzhen, China

Jul. 2025 — Aug. 2025

- Courses: Health Medical Big Data and Intelligent Statistics

Research & Work Experience

MacroStat (China) Clinical Research Co., Ltd. (Tigermed Group)

STATISTICIAN I

Shanghai, China

Jul. 2021 — Mar. 2022

- Supported clinical trial statistical analyses using SAS programming and standard statistical methods

Publications

REPRESENTATIVE JOURNAL ARTICLES (* CO-FIRST AUTHOR; # CORRESPONDING AUTHOR)

1. **Z. Lu**, L. Fu, L. Yang, T. Tian, Y. Gao, X. Meng[#], and H. Zou[#]. "Hesitancy to Undergo SARS-CoV-2 Rapid Antigen Testing in China: Nationwide Cross-sectional Study". *JMIR Public Health Surveill* 9 (2023), e43555. doi: [10.2196/43555](#). (Clarivate's 2022 JCR: Q1, JIF = 14.557; latest JCR data available at publication).
2. **Z. Lu**^{*}, B. Dong^{*}, P. Yao, Y. Huang, L. Fu, B. Wang, X. Huang, X. Mao, S. Xu, S. Liu[#], P. Sun[#], and H. Zou[#]. "Characteristics of Human Papillomavirus Prevalence and Infection Patterns Among Women Aged 35—65 in Fujian Province, China: A Nine-Year Retrospective Observational Study". *Journal of Medical Virology* 97.2 (2025), e70238. doi: [10.1002/jmv.70238](#). (2024 JCR: Q1, JIF = 6.8).
3. **Z. Lu**^{*}, J. Zou^{*}, H. Zhang^{*}, M. Zou^{*}, Y. Fu^{*}, R. Zhang, H. Shi, W. Wu, B. Xue, R. Wang, X. Yang, J. Cai, L. Gan, S. Liu, Y. Cai, Z. Peng[#], J. Li[#], L. Yang[#], J. Chen[#], and H. Zou[#]. "A machine learning model for predicting adverse prognostic events in patients with neurosyphilis: results from the DEFEAT-NS study". *iScience* (2026), p. 115104. doi: [10.1016/j.isci.2026.115104](#). (2025 JCR: Q1, JIF = 4.1).
4. **Z. Lu**^{*}, B. Dong^{*}, H. Cai^{*}, T. Tian^{*}, J. Wang^{*}, L. Fu, B. Wang, W. Zhang, S. Lin, X. Tuo, J. Wang, T. Yang, X. Huang, Z. Zheng, H. Xue, S. Xu, S. Liu[#], P. Sun[#], and H. Zou[#]. "Identifying Data-Driven Clinical Subgroups for Cervical Cancer Prevention With Machine Learning: Population-Based, External, and Diagnostic Validation Study". *JMIR Public Health Surveill* 11 (2025), e67840. doi: [10.2196/67840](#). (2024 JCR: Q1, JIF = 3.5).
5. H. Zou^{*}, **Z. Lu**^{*}, W. Weng^{*}, L. Yang^{*}, L. Yang^{*}, X. Leng, J. Wang, Y.-F. Lin, J. Wu, L. Fu, X. Zhang, Y. Li, L. Wang, X. Wu, X. Zhou, T. Tian, L. Huang, C. M. Marra[#], B. Yang[#], T.-C. Yang[#], and W. Ke[#]. "Diagnosis of neurosyphilis in HIV-negative patients with syphilis: development, validation, and clinical utility of a suite of machine learning models". *eClinicalMedicine* 62 (2023), p. 102080. doi: [10.1016/j.eclinm.2023.102080](#). (2023 JCR: Q1, JIF = 15.1).
6. L. Yang^{*}, **Z. Lu**^{*}, J. Bian, F. Li, and H. Zou[#]. "Association between chronic viral infection-related hospitalization and risk of cardiovascular disease: A population-based cohort study". *Journal of Medical Virology* 96.1 (2024), e29350. doi: [10.1002/jmv.29350](#). (2023 JCR: Q1, JIF = 12.7; Cover image based on this article).

7. B. Dong*, **Z. Lu***, T. Yang*, J. Wang, Y. Zhang, X. Tuo, J. Wang, S. Lin, H. Cai, H. Cheng, X. Cao, X. Huang, Z. Zheng, C. Miao, Y. Wang, H. Xue, S. Xu, X. Liu, H. Zou#, and P. Sun#. "Development, validation, and clinical application of a machine learning model for risk stratification and management of cervical cancer screening based on full-genotyping hrHPV test (SMART-HPV): a modelling study". *The Lancet Regional Health —Western Pacific* 55 (2025). doi: [10.1016/j.lanwpc.2025.101480](https://doi.org/10.1016/j.lanwpc.2025.101480). (2024 JCR: Q1, JIF = 7.6; Clarivate Highly Cited Paper).

SUBMITTED MANUSCRIPTS

1. **Z. Lu**, Y. Liu, S. Du, Z. Fan, Z. Wan, S. Huang, Z. Huang, H. Zou, and S. Liu#. "Whole-genome sequencing and multi-omic Mendelian randomization reveal genetic and proteometabolic determinants of cervical preinvasive and invasive disease" (Submitted to *Genome Medicine*).

SOFTWARE PACKAGES

1. **Z. Lu**. "omixvizpy: A Python package for omics data visualization". *Software* (2025). URL: <https://pypi.org/project/omixvizpy>. (Python package).
2. **Z. Lu**. "omixVizR: A Toolkit for Omics Data Visualization". *Software* (2025). DOI: [10.32614/CRAN.package.omixVizR](https://doi.org/10.32614/CRAN.package.omixVizR). (R package).
3. **Z. Lu**. "lulab.utils: Supporting Functions Maintained by Zhen Lu". *Software* (2024). DOI: [10.32614/CRAN.package.lulab.utils](https://doi.org/10.32614/CRAN.package.lulab.utils). (R package).

Patents

2023 Non-therapeutic clinical diagnosis system for neurological syphilis based on machine learning

LU504466 (rank 1/1)

Honors & Awards

Sep. 2025	Doctoral National Scholarship	Ministry of Education, China
Sep. 2025	First Prize Scholarship	Sun Yat-sen University
Sep. 2025	Outstanding Teaching Assistant	Sun Yat-sen University
Nov. 2024	Doctoral National Scholarship	Ministry of Education, China
Sep. 2024	Presidential Scholarship	Sun Yat-sen University
Sep. 2023	First Prize Scholarship	Sun Yat-sen University

Teaching Experience

2022, 2024	Undergraduate Teaching Assistant in Epidemiology	Sun Yat-sen University
2023	Undergraduate Teaching Assistant in English for Public Health	Sun Yat-sen University

Service & Outreach

WeChat Official Account: 陆震生物统计 (Lu Zhen Biostatistics; @luzhen-biostat; ~9,100 subscribers)

FOUNDER & LEAD MAINTAINER

- Created educational content on biostatistics, artificial intelligence, and statistical programming

OpenBiostat GitHub Organization

FOUNDER & LEAD MAINTAINER

- Developed and maintained open-source resources for biostatistics
- Founded Biostat Weekly, a weekly newsletter highlighting recent research, tools, and resources in biostatistics

Academic Advising

MENTORSHIP & CONSULTATION

- Mentored graduate and undergraduate students on statistical analysis and research writing
- Provided [consultation](#) in biostatistics, epidemiology, and academic writing

Academic Reviewer

PEER REVIEWER

- Reviewed manuscripts and provided peer-review reports for academic journals

Skills

Programming & Research Tools	SAS, SQL, R, Python, Bash/Shell; Git, Markdown, Quarto, LaTeX
Software & Platforms	Visual Studio Code, SAS Enterprise Guide, RStudio, GitHub, Overleaf, PASS
Languages	Mandarin Chinese (native), English (fluent)